

The Attralucian Essays:

Exploring the Finite



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The Attralucian Essays



The Name Is Not the Mathematics:
Language, Compression, and the Problem
of Mathematical Naming

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Overview

Mathematics presents itself as a language of unusual precision, yet its vocabulary is filled with inherited names: Abelian, Euclidean, Cartesian, Gaussian, Hermitian, Noetherian, Hamiltonian, Lagrangian, Bayesian, Markovian, and many others. To the initiated mathematician these terms are efficient; they appear to identify familiar structures almost immediately. To someone outside the established linguistic community, however, many of them reveal almost nothing of the operations, relations, or procedures they supposedly describe.

This essay begins from the simple observation that a term such as *Abelian* is not itself mathematical content. It is a historical name functioning as a highly compressed pointer into a previously learned network of meanings. From this starting point, mathematical naming can be examined as a linguistic process in which procedures become relations, relations acquire names, and names eventually come to appear as properties of mathematical objects. Arthur F. Bentley's *Linguistic Analysis of Mathematics* provides an intriguing historical point of contact with this problem. The present essay does not attempt a full analysis of Bentley's work, but it establishes a trajectory from which such an analysis can later be connected more formally with Finite Symbolic Mechanics, Functional Symbolic Trajectories, Finite Symbolic Logic,

and related developments.

The Name

There is something curious about the way mathematicians name things. A particularly clear example is the word *Abelian*.

Within mathematics the term is completely ordinary. An Abelian group is a group whose operation is commutative. For suitable elements a and b , one writes

$$a \circ b = b \circ a.$$

To somebody already trained in mathematics, the word *Abelian* is efficient. A considerable amount of information appears to arrive with eight letters. But the letters themselves contain almost none of that information.

There is nothing about the word *Abelian* that tells a reader that two operations may be performed in either order while preserving the specified relation. The word does not describe the operation, identify the elements, state the rule, or explain what is being compared. It points to Abel. More precisely, it points into a historical and linguistic community within which the name *Abel* has acquired a particular mathematical trajectory.

For the initiated reader,

Abelian

can activate something resembling

Abel $\rightarrow$ group theory $\rightarrow$ commutativity $\rightarrow a \circ b = b \circ a$.

For another reader, the same word may activate almost nothing.

This makes *Abelian* interesting not because it is a poor mathematical term, but because it exposes something that normally remains hidden. It reveals how much mathematical language depends upon previously stabilised linguistic trajectories. The name is not the mathematics. It is a pointer to the mathematics.

The Mathematical Club

Specialised communities inevitably develop specialised languages. Medicine, engineering, law, philosophy, computing and physics all compress long descriptions into shorter expressions, and mathematics is no exception.

The advantage is obvious. Once a community has learned the required language, enormously complicated relations can be communicated economically. The price of this economy is less often noticed. Terms begin to function

as code words.

The expert sees

Abelian group

and reconstructs an extensive body of assumptions, definitions and allowable operations. The novice sees an unfamiliar adjective attached to an unfamiliar noun. The difference does not lie in the marks on the page; the marks are identical. The difference lies in the trajectory available to the reader.

This raises an interesting question. Where, exactly, does the mathematical meaning reside? It cannot reside entirely in the printed word *Abelian*, because somebody unfamiliar with the term cannot extract the mathematical structure merely by inspecting its letters. Nor can it reside entirely in the accompanying symbols, because symbols themselves require learned conventions. The signs

$=, +, \in, \forall, \circ$

do not announce their permitted uses independently of a linguistic and mathematical community.

Meaning therefore seems to depend upon something larger than the isolated symbol. It depends upon the relationships through which the symbol has been learned and

used.

From Person to Property

Eponymous mathematics makes this particularly easy to see. Consider the trajectory:

Abel $\rightarrow$ Abelian.

A person's name becomes an adjective. The adjective then attaches to a mathematical noun:

Abelian group.

With sufficient use, the adjective begins to behave almost like the name of an intrinsic property:

G is Abelian.

The transformation is linguistically remarkable. A historical person becomes a name. The name becomes a modifier. The modifier becomes a classification. The classification is eventually treated as though it describes a property residing within a mathematical entity.

The same broad movement can be seen repeatedly:

Euclid $\rightarrow$ Euclidean,

Gauss $\rightarrow$ Gaussian,

Hilbert $\rightarrow$ Hilbert space,

Hamilton $\rightarrow$ Hamiltonian,

Lagrange $\rightarrow$ Lagrangian,

Noether $\rightarrow$ Noetherian.

To an established mathematician these terms cease to feel particularly strange. Their historical origins fade behind repeated use. That fading is itself significant. The trajectory by which the term acquired its meaning gradually disappears, leaving behind what appears to be a stable mathematical object or property.

One might schematically write:

procedure $\rightarrow$ repeatable relation $\rightarrow$ name $\rightarrow$ property $\rightarrow$ apparent object

The linguistic history becomes compressed. Eventually the compression becomes so successful that it is difficult to see that compression has taken place at all.

Unpacking the Compression

The word *Abelian* can therefore be treated as a compressed representation. Unpacking it produces something much less mysterious.

We require some collection of admissible symbols or elements. We require a specified operation. We perform the operation on admissible pairs. We then reverse the order of the operands and compare the registered results. Schematically,

$$G \rightarrow \{a, b, \dots\} \rightarrow \circ \rightarrow a \circ b \rightarrow b \circ a \rightarrow \text{comparison}.$$

The mathematical content lies in this network of operations and relations. *Abelian* merely provides a convenient address.

This suggests an important distinction. A name can be operationally useful without being foundationally explanatory. One might express the distinction simply:

An eponym may be an admissible label, but
it is not an admissible explanation.

There is no practical reason to abolish the word *Abelian*. On the contrary, once its trajectory is shared it is an extremely efficient linguistic device. The difficulty begins only when the compression is mistaken for the thing being compressed.

What Remains When the Name Is Removed?

A useful test follows. Remove the name. What remains?

If we remove *Abelian*, the operational relation remains available. Instead of saying

G is Abelian,

we can specify what must be done and what relation must hold when the order is changed. The name is dispensable. The operational trajectory is not.

This reversal is worth dwelling upon because mathematical education usually proceeds in the opposite direction. A term is introduced, given a definition, and subsequently used as though the term carries the definition within itself. Initially,

Abelian $\rightarrow$ definition.

After repeated use, however, the trajectory is psychologically reversed:

definition $\rightarrow$ Abelian.

The compressed label becomes sufficient for an expert reader. This is extraordinarily powerful. It is one of the reasons mathematics can construct long symbolic arguments without repeatedly expanding every earlier definition. But the same efficiency can conceal the structure through which meaning is being carried.

Mathematics and Its Surrounding Language

This leads naturally towards Arthur F. Bentley's *Linguistic Analysis of Mathematics*, published in 1932. Bentley is of interest because he explicitly turned attention towards the linguistic conditions surrounding mathematical symbolism. His concern was not merely with isolated mathematical notation, but with what might be described as the language in which symbols are introduced, interpreted, related and communicated.

At first sight it is tempting to divide mathematical discourse into two regions:

ordinary language | mathematical notation.

On one side stand words such as *group*, *point*, *space*, *therefore*, *every*, *exists*, and *Abelian*. On the other stand

$x, \quad +, \quad =, \quad \forall, \quad \exists, \quad \in .$

But the boundary is much less obvious than it first appears. The mathematical symbols themselves must be learned. Their permissible operations must be taught. Their meanings change historically. Their uses depend upon convention. Their interpretation depends upon surrounding symbols.

The sign $=$, for example, cannot function merely by being present on a page. A reader has to possess a trajectory of use that determines what sort of relation is being asserted and under which rules transformations involving that relation are permitted. The same is true of $+$, 0 , 1 , $\in$, or any other mathematical mark.

The apparent distinction between linguistic symbols and mathematical symbols may therefore conceal a deeper continuity. Both are symbols. Both obtain their practical meaning through use. Both depend upon established relations. Both participate in historically developed systems of interpretation.

Is Mathematics Embedded in Language?

Bentley's work raises a question that deserves careful examination. Is mathematics something separate that is subsequently embedded within language?

One conventional picture might be represented as

mathematical structure $\rightarrow$ symbolic notation $\rightarrow$ linguistic explanation.

According to such a picture, the mathematical structure exists independently, while language is merely employed to communicate it. But another possibility is available. Mathematics may itself be a specialised form of symbolic language. That would give something closer to

symbolic distinction $\rightarrow$ repeatable operation $\rightarrow$ stable convention $\rightarrow$ m

Under this interpretation, mathematics does not escape the problems of language merely by adopting specialist notation. Its language is more tightly constrained, but it remains language. The symbols are not free of history. They are not free of agreed rules. They are not free of interpretation. And they are certainly not free of

compression.

The Appearance of Mathematical Things

The problem extends beyond eponyms. Indeed, the eponyms may be the easiest cases precisely because their linguistic origins remain visible. We can still see the person inside *Abelian*. But what of words such as

point, line, set, space, field, function, dimension

These no longer appear to us as eponyms. Their histories are buried much more deeply. They appear natural. A point seems to be a thing. A set seems to be a thing. A function seems to be a thing. A number seems to be a thing. Yet each is introduced through linguistic and symbolic operations.

A definition is established. Permitted manipulations are specified. Examples are given. Relations are repeated. A community stabilises usage. Eventually the term begins to appear self-evident. The word ceases to look like a linguistic construction and starts to look like the name of something that was waiting to be discovered. This transformation may be one of the most important effects

of mathematical language.

From Operation to Noun

The problem connects with a much wider tendency within language. Processes are repeatedly converted into nouns. Relations become things. Activities become entities. Regularities become properties. Mathematics intensifies this tendency because it requires stable handles that can be reused within extended symbolic constructions.

Suppose a family of operations repeatedly exhibits some identifiable relation. The relation is named. The name can then be manipulated without continually repeating the underlying procedure. Schematically,

many operational instances $\rightarrow$ recognised relation $\rightarrow$ name.

The name can subsequently enter new symbolic constructions:

name $\rightarrow$ classification $\rightarrow$ new relations.

This is an extraordinarily effective form of compression. But it may also encourage us to attribute an existence to the compressed symbol that originally belonged only to the operations through which the symbol was produced.

The noun begins to dominate the process.

Compression Rather Than Abstraction

This offers another way of approaching what is commonly called mathematical abstraction. The word *abstraction* often suggests that some property has been separated from particular instances and has somehow acquired an independent mathematical existence. But much of what is described as abstraction may instead be understood more cautiously as symbolic compression.

Suppose several different operational systems exhibit a sufficiently similar registered relation:

$$A_1, \quad A_2, \quad A_3, \quad \dots, \quad A_n.$$

Rather than continually restating the operational similarity, we assign a symbolic label:

$$A_1, \dots, A_n \rightarrow L.$$

The new symbol L allows the previously identified relation to be carried economically into subsequent reasoning. Nothing mysterious need have been extracted from the systems. Nothing need have entered an abstract

realm. A finite symbolic community has recognised a repeatable relation and compressed it into another symbol. The new symbol then acquires its own trajectory.

From this perspective, the movement may be written as

registered relations $\rightarrow$ classification $\rightarrow$ symbolic compression $\rightarrow$ new sy

What is ordinarily called abstraction may therefore contain a substantial and sometimes overlooked linguistic component.

The Reader Reconstructs the Mathematics

This also shifts attention towards the reader. A mathematical expression does not deliver an identical meaning to every observer. The reader reconstructs its meaning using previously learned symbolic relations.

Consider again the word

Abelian.

For one reader it may activate almost immediately a large interconnected body of mathematics. For another it remains merely an unfamiliar mark. The physical inscrip-

tion has not changed. What differs is the available history of interaction with that inscription and with the surrounding symbolic system.

Meaning therefore appears less like a substance contained inside a symbol and more like a trajectory capable of being reactivated. The experienced mathematician does not inspect each symbol anew. A previously established path is followed. The trajectory has become sufficiently stable that the intermediate stages largely disappear from conscious view.

This may explain something of the peculiar density of advanced mathematics. A short expression can activate an enormous network of prior relations, but only for somebody whose previous symbolic history permits the reconstruction. The page is therefore not carrying all of the mathematics. Part of the mathematics is being supplied by the reader.

Towards Finite Symbolic Mechanics

This observation begins to approach Finite Symbolic Mechanics, but at this stage it is useful not to force the connection too quickly. The immediate question is simpler. What is actually present when mathematics is performed, written, communicated, or measured?

There are finite marks. There are distinctions between marks. There are rules governing how marks may be combined. There are histories through which those rules have been established. There are communities that recognise particular operations as admissible. There are readers who reconstruct trajectories from finite inscriptions.

From an FSM perspective, this suggests that the name itself cannot provide foundational admissibility. The relevant question becomes:

What finite symbolic trajectory allows the named relation to be produced, distinguished, registered and subsequently recognised?

The word *Abelian*, considered alone, cannot answer this. It identifies neither the registration nor the operation. It is instead a compressed pointer towards a trajectory already stabilised elsewhere. This does not make it useless. It locates its usefulness.

A Possible Reversal

Traditional mathematical discourse often proceeds by naming and then manipulating named objects. FSM suggests that it may sometimes be valuable to run the trajectory backwards. Instead of

name $\rightarrow$ property $\rightarrow$ mathematical object,

one might ask for

symbolic object $\rightarrow$ permitted operations $\rightarrow$ registered relations $\rightarrow$ nam

The second form preserves the provenance of the name. The noun remains attached to the process that produced it. The label remains visibly a label.

This may become particularly important when mathematics is applied to measurement, where finite distinctions must cross between physical instrumentation and symbolic representation. At that point, the question of admissibility cannot be answered solely by appealing to a previously agreed mathematical noun. The route by which the symbolic state was produced matters. Its provenance matters. Its uncertainty matters. The distinction by which one state was separated from another matters. The mathematics can no longer simply begin after these questions have been discarded.

Functional Symbolic Trajectories

This provides one possible later bridge to the concept of the Functional Symbolic Trajectory. The term *Abelian* can be considered not as a container holding a mathematical meaning but as a node capable of reactivating a learned trajectory. Very approximately,

“Abelian” $\rightarrow$ prior language $\rightarrow$ definitions $\rightarrow$ operations $\rightarrow$ relations -

The apparent stability of the word arises from the repeatability of this reconstruction within a sufficiently trained community. This would permit mathematical meaning to be examined dynamically rather than as a collection of isolated symbolic objects.

The symbol becomes part of a path. The path has history. The path can be compressed. The path can bifurcate. The path can fail to reconstruct in another reader. And the same visible symbol may therefore participate in very different functional trajectories. That possibility will require much more careful development.

Towards a Linguistic Analysis of Mathematics

Bentley’s work may provide an important historical bridge for precisely this reason. The task, however, should not initially be to decide whether Bentley was right or wrong, nor to recruit his arguments prematurely into Finite Symbolic Mechanics. A more productive question may be:

Where does Bentley believe language ends and mathematics begins?

If he preserves such a division, the boundary itself becomes an object of investigation. If he does not preserve it, then the manner in which he understands mathematical symbolism becomes still more interesting.

His work was produced long before contemporary neural networks, information theory as presently understood, modern digital computation, or the present framework of Finite Symbolic Mechanics. He therefore approached the linguistic problem from a very different historical position. That difference is potentially valuable. It provides an independent trajectory.

Reading Bentley may reveal earlier attempts to identify the dependence of mathematical practice upon language without importing the conceptual vocabulary developed here. Only after that reading will it become possible to establish precisely where his trajectory converges with, crosses, or departs from the present one.

The Name Is Not the Mathematics

The original observation can now be stated more carefully. Mathematicians use names extensively because names are powerful devices of symbolic compression. The practice is not peculiar to mathematics, nor is it inherently problematic. The interesting point is that success-

ful compression gradually conceals itself.

The expert stops seeing the historical name. The term becomes a property. The property becomes associated with an object. The object appears independent of the linguistic trajectory through which it became available.

The eponym offers us a rare opportunity to see the mechanism before the join completely disappears. *Abelian* still visibly contains Abel. It therefore reminds us that a mathematical term can carry historical provenance even while functioning as apparently timeless mathematical vocabulary.

The same question can then be carried further. What provenance is hidden inside *number*? What operations are hidden inside *equality*? What trajectories are compressed inside *point*, *set*, *space*, or *dimension*? At what stage does a repeatable symbolic operation become a noun? At what stage does the noun come to appear as an entity? And at what stage do we forget that the entity began as a relation performed through symbols?

These questions extend well beyond terminology. They concern the foundations of mathematical representation itself. For the moment, however, a useful principle can be retained:

The name may preserve the history of a mathematical idea, compress its operational meaning, and permit efficient communication within

a community. But the name is not the operation, the relation, or the registration.

The name is not the mathematics.

The next step is therefore not to discard the name. It is to unfold it.

A Note Towards Further Development

This essay is intended as a preliminary linguistic investigation rather than a completed formal argument. A subsequent study of Arthur F. Bentley's *Linguistic Analysis of Mathematics* may provide an historical and conceptual bridge from which the analysis can be developed more rigorously.

Several trajectories are already visible. One leads towards Finite Symbolic Mechanics and the requirement that symbolic claims retain their admissible operations, provenance and conditions of registration. Another leads towards Functional Symbolic Trajectories, in which mathematical terms can be studied as nodes within reconstructable histories of symbolic use rather than as isolated containers of meaning. A third leads towards Finite Symbolic Logic, where terms such as identity, equality, implication and admissibility can be reconsidered not merely as inherited logical primitives but as finite sym-

bolic operations.

These developments should follow the historical investigation rather than precede it. Bentley's text can therefore be read not as an authority to be absorbed into the present framework, but as another trajectory with which the present one can interact.

The point of contact is already clear. Mathematics does not merely use language to describe itself. Mathematics is performed through symbols whose meanings depend upon relations, histories, conventions, operations and readers. Once this is recognised, the linguistic analysis of mathematics becomes inseparable from the analysis of mathematics itself.

The Measurement This Essay Makes

This essay has not attempted to replace mathematical names, nor to found a new vocabulary. It has attempted a more limited act: a measurement.

The quantity under investigation was not Abel, nor the group, nor commutativity conceived as an inhabitant of an object. It was the interval between a name and the finite procedure that the name is later taken to contain. The instrument was comparison. One side of the comparison is the compressed term

Abelian.

The other side is the operational trajectory that remains when the term is removed: admissible elements, a specified operation, a reversal of order, and a registered relation between the two results.

What the comparison distinguishes is therefore not a person from a theorem. It distinguishes a label from a measurement. The equality

$$a \circ b = b \circ a$$

is already a recorded distinction. It says that two finite productions, performed in opposite order, do not separate under the criterion in use. The word *Abelian* compresses that recorded distinction so that it can be carried forward without repeating the procedure. The compression is admissible. It becomes misleading only when the record is treated as an intrinsic property that no longer requires conditions of production.

The same measurement can now be stated of the essay itself.

A finite inscription has been used to separate three things that ordinary mathematical speech often presents as one: the historical name, the operational relation, and the apparent object that later seems to bear the name as a prop-

erty. The result of the measurement is modest. The name is recoverable as provenance. The relation is recoverable as a repeatable comparison. The object-appearance is recoverable as the success of compression.

What remains unmeasured here is equally important. This essay does not determine the full provenance of terms such as *number*, *set*, or *equality*. It does not assign a bound to the uncertainty of expert reconstruction. It does not yet exhibit, for each inherited noun, the finite trajectory by which the noun became available. Those are later measurements. They require other instruments: historical reading in the case of Bentley, and an explicit account of admissible operations, registration and uncertainty in the case of Finite Symbolic Mechanics.

The present result is only this. Once a mathematical name is treated as a compressed record of a measurement, rather than as the thing measured, the name can be retained without being mistaken for the mathematics. The measurement does not discard the label. It restores the conditions under which the label was produced.

Bibliography

- [1] Arthur F. Bentley, *Linguistic Analysis of Mathematics*, Bloomington, Indiana, 1932.